

Paper 1: Basic

1. Simplify.

$3(2x - 1) - 2$

[1]

[N22/I/2(b)]

2. (a) Factorise completely
- $30 - 45a$
- .

[1]

(b) Expand and simplify $(5x - 4y)^2$.

[2]

[N22/I/11]

3. Factorise
- $4x^2 + 4x - 15$
- .

[2]

[N22/I/16]

4. Write as a single fraction in its simplest form
- $\frac{3}{2x-3} - \frac{2}{3x-2}$
- .

[3]

[N22/I/21]

5. Write as a single fraction in its simplest form
- $\frac{3x}{4} - \frac{2(x-4)}{3}$
- .

[2]

[N21/I/5]

6. Simplify
- $8a - 5b - 3(2a + 3b)$
- .

[2]

[N21/I/9(b)]

7. The formula
- $s = 3.5(t - 2)^2$
- can be used to calculate the distance,
- s
- metres, travelled by a cyclist in a time of
- t
- seconds, where
- $t > 2$
- .

Find t when $s = 14$.

[1]

[N21/I/16(a)]

8. Factorise completely
- $6x^2 - y + 3xy - 2x$
- .

[2]

[N21/I/21(a)]

9. Simplify
- $\frac{4a^2}{3b} \div \frac{10ab}{21}$
- .

[2]

[N20/I/5(b)]

- 10.
- $y = 16 - 4x^2$

(a) Find y when $x = -2$.

[1]

(b) Rearrange the formula to make x the subject.

[2]

[N20/I/13]

11. (a) Simplify
- $3(3x + 2y) - 5(x - 3y)$
- .

[2]

(b) Factorise completely $12ab - 9ax - 8by + 6xy$.

[2]

[N20/I/19]

12. Write as a single fraction in its simplest form
- $\frac{1}{2x-3} - \frac{3}{3x-1}$
- .

[2]

[N19/I/6]

13. (a) Factorise $x^2 - y^4$. [1]
 (b) Factorise completely $6ab + 1 - 3a - 2b$. [2]
 [N19/I/11]
14. Show that $(3n - 1)^2 + 2$ is a multiple of 3 for all integer values of n . [2]
 [N18/I/5]
15. Factorise completely
 (a) $8p^2q - 6pq^3$, [2]
 (b) $6x^2y - 2xy + 3x - 1$. [2]
 [N18/I/13]
16. (a) Simplify $3(2x - 1) + 1$. [1]
 (b) Factorise $6x + 18xy$. [1]
 [N16/I/1]
17. Factorise completely $4ax + 12by - 16ay - 3bx$. [2]
 [N16/I/2]
18. n is a positive integer.
 Show that, for all n , $(5n + 1)^2 - (5n - 1)^2$ is a multiple of 20. [2]
 [N16/I/4]
19. Write as a single fraction in its simplest form $\frac{3}{(x - 4)^2} - \frac{1}{(4 - x)}$. [2]
 [N16/I/9]
20. One day, John used this formula to estimate the air temperature at different heights above sea level.
- $$T = 21 - \frac{h}{120},$$
- where T is the temperature in degrees Celsius,
 and h is the height, in metres, above sea level.
- (a) A light aircraft records the air temperature as 0°C .
 Use the formula to estimate the height of the aircraft above sea level. [1]
- (b) Two jet aircraft are flying at different heights.
 The difference in the temperatures at their two heights is 25°C .
 Estimate the difference between the heights of the two aircraft. [2]
 [N16/I/13]
21. Simplify $\frac{4x}{3} - \frac{3(2 - 5x)}{4}$. [2]
 [N15/I/5]
22. Factorise fully $4ax - 3ay - 8bx + 6by$. [2]
 [N15/I/7]

1. Simplify $\frac{4x^2 - 8ax - 3x + 6a}{3x^2 - 12a^2}$. [3]
[N22/II/1(d)]

2. It is given that $a = 2n(n - 3k)$.
 - (i) Find a when $n = 0.5$ and $k = -1.2$. [1]
 - (ii) Express k in terms of a and n . [2]
[N21/II/1(b)]

3. (a) Express as a single fraction in its simplest form $\frac{x}{(3 - 2x)^2} - \frac{5}{3 - 2x}$. [2]
- (b) Simplify $\frac{4x^2 - 16}{3x^2 + x - 10}$. [3]
[N20/II/1(c), (e)]

4. (a) Simplify $\frac{4p^2r}{3} \div \frac{2r^3}{p}$. [1]
- (b) $a = \frac{3b + 4c}{5 - b}$
 - (i) Evaluate a when $b = 6$ and $c = -2$. [1]
 - (ii) Express b in terms of a and c . [2]
[N19/II/1(a), (b)]

5. (a) Write as a single fraction in its simplest form
 - (i) $\frac{5t^2}{v} \div \frac{25t}{v^3}$, [1]
 - (ii) $\frac{4}{3 - 2y} - \frac{5}{y + 3}$. [2]
- (b) Simplify $\frac{16x^2 - 9}{4x^2 - 9x - 9}$. [3]
[N18/II/1(a), (b)]

6. (a) Express as a single fraction in its simplest form $\frac{2y}{5 - 2y} - \frac{3y}{(5 - 2y)^2}$. [2]
- (b) Simplify $\frac{18h^3j^3}{5k^3} \div \frac{3h^5k}{10j^2}$. [2]
[N17/II/1(b), (c)]

7. (a) It is given that $a = \frac{4b - 5c}{b + c}$.
 - (i) Find a when $b = 5$ and $c = -2$. [1]
 - (ii) Express b in terms of a and c . [2]
- (b) Simplify $\frac{9x^2 - 4}{3x^2 - 10x - 8}$. [3]
[N16/II/1(a), (d)]

8. (a) Factorise $9x^2 - 16y^2$. [1]
- (b) Express as a single fraction in its simplest form
 - (i) $\frac{15xy}{12} \div \frac{9x^2}{4y}$, [1]
 - (ii) $\frac{6}{2x - 3} - \frac{1}{x + 2}$. [2]
[N15/II/1(a), (b)]

9. (a) The diagram shows part of a number grid.

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33			

A rectangular column outlining four numbers, as shown, can be placed anywhere on the grid.

- (i) If t represents the number in the top of the rectangle, find an expression, in terms of t , for the product of the top number and the bottom number in the column. [1]
 - (ii) Explain why the difference between the product of the middle two numbers and the product of the top and bottom numbers is always 72. [2]
 - (iii) The rectangular column is placed on the grid such that the sum of the four numbers outlined is 360. Find the value of the largest number in the column. [3]
- (b) In a sequence, S , the difference between consecutive terms is constant.
- The third term of this sequence is 36.
- The sixth term of this sequence is 60.
- (i) Find an expression, in terms of n , for the n th term of this sequence. [2]
 - (ii) Explain why the terms of the sequence are all multiples of 4. [1]

[N19/II/5]

10. The first four terms in a sequence of numbers are given below.

$$T_1 = 2^2 + 5 = 9$$

$$T_2 = 3^2 + 8 = 17$$

$$T_3 = 4^2 + 11 = 27$$

$$T_4 = 5^2 + 14 = 39$$

- (a) Find T_5 . [1]
- (b) Explain why the value of T_n must be odd for all values of n . [1]
- (c) Show that the n th term of the sequence, T_n , is given by $n^2 + 5n + 3$. [3]
- (d) T_p and T_{p+1} are consecutive terms in the sequence. Find and simplify an expression, in terms of p , for $T_{p+1} - T_p$. [3]
- (e) Explain why two consecutive terms of the sequence cannot have a difference of 4. [1]

[N16/II/4]

11. (a) The first four terms in a sequence are 55, 51, 47 and 43.

(i) Find an expression, in terms of n , for the n th term, T_n , of this sequence. [2]

(ii) Evaluate T_{25} . [1]

- (b) The diagram shows part of a number grid.

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40

A square outlining four numbers, as shown, can be placed anywhere on the grid.

(i) If n represents the number in the top left corner of the square, write down an expression, in terms of n , for the number in the bottom right corner of the square. [1]

(ii) Show that the difference between the products of the numbers in the opposite corners of the square is always 8. [2]

(iii) Show that the sum of the four numbers in the square cannot be 260. [3]

[N15/II/8]

Paper 2: Advanced

12. Here are the first four terms of a sequence.

$$\frac{1}{2}, \quad \frac{5}{4}, \quad \frac{9}{6}, \quad \frac{13}{8}$$

(a) Find the fifth term of the sequence. [1]

(b) T_n is the n th term of the sequence.

Find an expression, in terms of n , for T_n . [3]

(c) The difference, D , between two consecutive terms of the sequence is $T_{n+1} - T_n$.

Show that $D = \frac{3}{2n(n+1)}$. [3]

(d) Explain why the difference between two consecutive terms of the sequence is always less than 1. [1]

[N21/II/7]